VERIFICATION CORE REPORT

Topic: **Spring Boot 3 & Java 21 Architecture** | College: **DPU**
Tech: **Java 21** | Professor: **Dr. Samruddhi Khade** | Deck ID: **test_e2e** | Date: 2026-10-17 10:24:15
Docker Image Digest: **eclipse-temurin:21-jdk-alpine@sha256:7b5d1e4392a874f6e21b03e2c34a6e8b7c439f018e652a129efb923415cf98a1**

Metric	Value	Metric	Value
Found vs Executed	3 found / 3 run	Unverified (Not Run)	0
Snippets Verified	1/3	Auto-Healed	1
Needs Review / Fail	0	Non-Deterministic	0
Exercises Proven	3/3	Mutation Score	9333%
Template QA Passed	5/5	Docker Backend Runs	0

1. Code Snippet Sandbox Verification

Slide	Language & Backend	Status	Duration	Captured Output
Slide 2	java 21 (isolated_local_subprocess) **[Wrapped: imports omitted]**	VERIFIED LOW ASSURANCE	557.6248440551758ms	`Spring Boot 3 Web Endpoint Initialized`
Slide 3	java 21 (isolated_local_subprocess)	AUTO HEALED	477.68640518188477ms	`Java 21 Virtual Thread Running`
Slide 4	java 21 (isolated_local_subprocess) **[Wrapped: imports omitted]**	VERIFIED LOW ASSURANCE	57.36810684204102ms	`Exercise Starter Code`

2. Exercise Proof & Mutation Score Matrix

Tier	Exercise Ref	Final Status	Attempts	Ref Sol	Naive Sol	Mutants Killed	Score
EASY	EX-2-EASY	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
MEDIUM	EX-2-MEDIUM	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
STRETCH	EX-4-STRETCH	VERIFIED	1	PASSED	FAILED (Good)	4/5	80.0%

EX-2-EASY (EASY) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner

EX-2-MEDIUM (MEDIUM) - Sandbox Execution Runs:

- **reference**: PASS
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner

EX-4-STRETCH (STRETCH) - Sandbox Execution Runs:

- **reference**: PASS
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-1**: PASS
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner

3. Template Compliance QA

Slide	Compliance Status	Details
Slide 1	COMPLIANT	All bounding boxes and fonts compliant.
Slide 2	COMPLIANT	All bounding boxes and fonts compliant.
Slide 3	COMPLIANT	All bounding boxes and fonts compliant.
Slide 4	COMPLIANT	All bounding boxes and fonts compliant.
Slide 5	COMPLIANT	All bounding boxes and fonts compliant.